

ELEMENT B: DOCUMENTATION AND ANALYSIS OF PRIOR DESIGN ATTEMPTS

PRIOR DESIGN ATTEMPT #1 – OBSTRUCTED FOOD SLED ENRICHMENT

Source: Personal interview with carnivore keeper Zookeeper A at a zoological facility.



Details: This enrichment device consists of a small circular sled placed on the ground with food hidden underneath. Zookeepers place sticks, rocks, or other objects on top of the sled, requiring

the leopard to first remove the obstructions and then flip the sled to access the food reward (Sloane). According to the keeper, this activity promotes problem-solving and physical interaction, particularly when used on slick surfaces in holding areas where the sled moves unpredictably.

Analysis: While this design encourages initial engagement and problem-solving, it does not fully meet the design requirements. The task becomes repetitive and predictable after repeated use, reducing long-term mental stimulation. Additionally, in the main enclosure, environmental features such as textured ground and structures make the challenge easier, decreasing its effectiveness. The bright, artificial appearance of the sled also conflicts with the requirement for a natural-looking enrichment item suitable for public display. Therefore, although the design is cost-effective and simple, it lacks durability in cognitive engagement and aesthetic integration.



**PRIOR DESIGN ATTEMPT #2 –
FLOATING WATERMELON
ENRICHMENT**

Source: “Enrichment.” Big Cat Rescue,
www.bigcatrescue.org/enrichment/.

Image: Details: This enrichment method involves placing a watermelon in water, allowing it to float. The leopard interacts by swatting, chasing, and

eventually retrieving the watermelon. After removing it from the water, the animal breaks it open to access the food inside (“Enrichment”). This activity promotes physical play and mimics aspects of prey interaction.

Analysis: Although engaging in the short term, this design does not meet the requirement for sustained cognitive enrichment. The task is quickly understood and completed, limiting repeated engagement. Additionally, it lacks complexity and does not require multi-step problem-solving. The enrichment is temporary and dependent on perishable materials, making it less practical for repeated or long-term use. While it appears natural, it fails to provide sufficient mental stimulation or durability.

PRIOR DESIGN ATTEMPT #3 – HANGING MEAT PUZZLE FEEDER

Source: Association of Zoos and Aquariums. “Animal Enrichment.” AZA, www.aza.org/animal-enrichment.



Image:

Details: This design involves suspending food such as meat from a rope or container, requiring the leopard to jump, pull, or manipulate the object to access the reward (Association of Zoos and Aquariums). Some variations include holes or compartments that require strategic interaction to release food, increasing engagement and physical activity.

Analysis: This design promotes physical exertion and some problem-solving; however, it may not meet size and safety constraints. Suspended systems require secure installation and may pose risks if materials fail or become entangled. Additionally, once the leopard learns the mechanism,

the challenge becomes repetitive. The design may also be difficult to scale to the required dimensions while maintaining effectiveness. While more stimulating than simpler enrichment, it lacks long-term variability and may not fully align with project constraints.

PRIOR DESIGN ATTEMPT #4 – SCENT-BASED ENRICHMENT OBJECTS

Source: Young, Robert J. Environmental Enrichment for Captive Animals. Wiley-Blackwell, 2003.



Image:

Details: Scent enrichment involves applying novel or prey-related scents such as spices, perfumes, or animal-based scents to objects within the enclosure. Leopards investigate these scents through sniffing, rubbing, and marking behaviors, stimulating natural instincts (Young).

Analysis: While scent enrichment encourages natural behaviors, it does not fully address the need for interactive, multi-step problem-solving. The stimulation is primarily sensory rather than cognitive or physical. Additionally, scents dissipate over time, requiring frequent reapplication, which reduces sustainability. This approach also does not inherently include a food reward, limiting prolonged engagement and motivation.

PRIOR DESIGN ATTEMPT #5 – PUZZLE BOX FEEDER

Source: Bloomsmith, Mollie A., et al. “Environmental Enrichment for Captive Animals.” *ILAR Journal*, vol. 40, no. 3, 1999, pp. 129–138.



Details: Puzzle box feeders are enclosed containers with openings or mechanisms that require the animal to manipulate parts of the structure to release food. These may include sliding panels, rotating components, or hidden compartments (Bloomsmith et al.). These devices are widely used to increase cognitive engagement in captive animals.

Analysis: This design aligns well with the goal of cognitive enrichment, as it encourages problem-solving and persistence. However, many existing designs are made of plastic or brightly colored materials, which do not meet the requirement for a natural appearance. Additionally, more complex mechanisms can increase cost and may exceed the project budget. Some designs may also be too large or difficult to clean and maintain. While effective in theory, modifications are needed to meet constraints related to size, cost, and aesthetics.